Programming Fundamentals
COC1071

Lecture No. 07

Last Lecture Review

- Repetition
- While Loop
- Increment and Decrement operators

What Will You Learn Today?

- Solution of Home Work Assignment
- Repetition – do-While Loops
- Do-while Loop Examples

Repetition (Loops)

- Every loop can be defined by three way:
  1. Starting Point
  2. Ending Point
  3. Sequence of Moving

While Loop

- Structure of while loop can be defined as:

```
while(condition)
{
   statement(s);
}
```

While Loop

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{
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```

Increment and Decrement operators

- **++** is unary increment operator which replaces `a=a+1` into `a++`
- **--** is unary decrement operator which replaces `a=a-1` into `a--`
- Used as pre or post fashion

Increment and Decrement operators

Operator	Called	Sample
++	Pre-increment	`++a`
++	Post-increment	`a++`
--	Pre-decrement	`--a`
--	Post-decrement	`a--`

Problem (Statement)

- In a company, there are deductions from the salary of the employees for a fund. The deductions rules are as follows:
  - If salary is less than Rs.10,000 then no deduction
  - If salary is equal to or more than Rs.10,000 and less than Rs.20,000 then deduct Rs.1,000 as fund
  - If salary is equal to or more than 20,000 then deduct 7 % of the salary for fund
- Input salary from user and after appropriate deduction show the net payable amount.

Home Work Assignment

Problem (Solution)

```
int salary ;
float deduction, netPayable ;
cout << "Please enter the salary : " ;
cin >> salary ;

switch ( salary / 10000 )
{
    case 0 : // salary is less than 10,000
        // In this case deduction is zero
        deduction = 0;
        netPayable = salary ;
        cout << "Net Payable(salary - deduction)= " ;
        cout << salary << " - " << deduction << "=" ;
        cout << netPayable;
        break ;
```

Problem (Solution)

```
    case 1 : //salary is in range 10,000 - 19,999
        deduction = 1000 ;
        netPayable = salary - deduction ;
         cout << "Net Payable(salary - deduction)= ";
        cout << salary << " - " << deduction << "=" ;
        cout << netPayable ;
        break ;

    default : // salary is 20,000 or more
        deduction = salary * 7 / 100 ;
        netPayable = salary - deduction ;
        cout << "Net Payable(salary - deduction)= " ;
        cout << salary << " - " << deduction << "=" ;
        cout << netPayable ;
} // End of switch statement
```

Problem (Output)

```
Please enter the salary : 9000
Net Payable(salary - deduction)=9000 - 0 = 9000
```

```
Please enter the salary : 19000
Net Payable(salary - deduction)=19000 - 1000 = 18000
```

```
Please enter the salary : 21000
Net Payable(salary - deduction)=21000 - 1470 = 19530
```

do-while Statement

- There may be certain situations when the body of while loop does not execute even a single time
- This occurs when the condition in while is false
- In while loop, the condition is tested first and the statements in the body are executed only when this condition is true

do-while Statement

- If the condition is false, then the control goes directly to the statement after the closed brace of the while loop
- In while structure, the loop can execute zero or more times
- There may be situations where it may need that some task must be performed at least once

do-while Statement

- The structure of do-while loop is as follows:

```
do
{
   statement(s) ;
}
while ( condition ) ;
```

do-while Statement

- The structure of do-while loop is as follows:

```
do
{
   statement(s) ;
}
while ( condition ) ;
```

Problem – 1 (Statement)

- Write a program which inputs from user a character between **a** to **z** and guess whether the character matches with a pre-defined letter or not. If it matches, then display a message "**Congratulations**". It gives the user five chances or tries to guess the character.

Problem – 1 (Solution)

```
int main() {
    int tryNo = 0 ;
    char guess ;
    do {
        cout << "Enter character(a-z) to guess:" ;
        cin >> guess ;
        //check the entered character for equality
        if (guess == 's') {
            cout << "Congratulations..." ;
            tryNo = 6 ;
        }
        else {
            tryNo++ ;
        }
    }
    while ( tryNo < 5 ) ;
}
```

Problem – 1 (Solution)

```
int main() {
    int tryNo = 0 ;
    char guess ;
    do {
        cout << "Enter character(a-z) to guess:";
        cin >> guess ;
        //check the entered character for equality
        if (guess == 's') {
            cout << "Congratulations...";
            break ;
        }
        else {
            tryNo = tryNo + 1;
        }
    }
    while (tryNo < 5);
}
```

Problem – 1 (Statement)

```
Enter character(a-z) to guess:m
Enter character(a-z) to guess:s
Congratulations...
```

```
Enter character(a-z) to guess:a
Enter character(a-z) to guess:b
Enter character(a-z) to guess:c
Enter character(a-z) to guess:d
Enter character(a-z) to guess:s
Congratulations...
```

Problem – 2 (Statement)

- Calculate the sum of numbers from **1** to **100** using do-while loop.

Problem – 2 (Solution)

```
int main()
{
    int number = 1 ;
    int sum = 0 ;
    do
    {
        sum = sum + number ;
        number++ ;
    }
    while ( number <= 100 ) ;

    cout << "The sum of first " << number - 1;
    cout << " number is:\t " << sum << endl;
}
```

```
The sum of first 100 number is:  5050
```

Problem – 3 (Statement)

- Calculate the sum of even numbers from **1** to **100** using do-while loop.

Problem – 3 (Statement)

```
int main()
{
    int number = 0 ;
    int sum = 0 ;
    do
    {
        if ( number%2 == 0 )
        {
            sum = sum + number ;
        }
    number++ ;
    } while ( number <= 100 ) ;
    cout << "The sum of first " << number ;
    cout << " even numbers is:\t" << sum << endl ;
}
```

```
The sum of first 100 even numbers is:  2450
```

Problem – 4 (Statement)

- Calculate the sum of odd numbers from **1** to **100** using do-while loop.

Problem – 4 (Statement)

```
int main()
{
    int number = 0 ;
    int sum = 0;
    do
    {
        if ( number%2 != 0 )
        {
            sum = sum + number ;
        }
    number++ ;
    } while ( number <= 100 ) ;
    cout << "The sum of first " << number ;
    cout << "odd numbers is:\t" << sum << endl ;
}
```

```
The sum of first 100 odd numbers is:  2500
```

Problem – 5 (Statement)

- Write the table of **5** using do-while loop.

Problem – 5 (Solution)

```
#include<iostream>
using namespace std;

int main()
{
    int counter = 1 ;

    do
    {
        cout<<"5 X "<<counter<<" = "<<5*counter;
        cout << endl ;

        counter++ ;
    }
    while ( counter <= 10 ) ;
}
```

Problem – 5 (Output)

```
5 X 1 = 5
5 X 2 = 10
5 X 3 = 15
5 X 4 = 20
5 X 5 = 25
5 X 6 = 30
5 X 7 = 35
5 X 8 = 40
5 X 9 = 45
5 X 10 = 50
```

Recap

- Solution of Home Work Assignment
- Repetition – do-While Loops
- Do-while Loop Examples

Questions